Computer Science & Engineering department

Computer Vision

CS3203

Circle ID: Writer Identification

Rohit Roy, Nithin Nayak, N Krishna Pawar
2301AI54, 2301AI16, 2301AI41

22nd April 2026

Contents

1 Introduction **2**
1.1 The Problem Statement 2

2 Dataset Description **2**
2.1 Source of the Data 2
2.2 Challenges in the Dataset 2
2.3 Data Analysis 3

3 Methodology / Approach **3**
3.1 AC-GAN Discriminator 3
3.2 Kinematic Pressure Map & The Random Forest Bouncer 4
3.3 Frequency Domain & Siamese Pyramidal Attention 4

4 Result & Analysis **5**

5 Challenges **6**

6 Future Improvements **6**

1 Introduction

1.1 The Problem Statement

The task was to identify who drew a circle using only a scanned image of a hand drawn circle. A circle looks simple, yet it contains subtle cues from human motor behavior. However, appearance is also affected by pen/ink, which makes writer identification challenging.

The Competition entirely focused on writer identification.

2 Dataset Description

2.1 Source of the Data

The dataset consisted of the following files:

- train.csv contains labels: image_id, image_path, writer_id, pen_id
- test.csv contains: image_id, image_path
- Image are provided as PNG files.

Label Format: writer_id is a string (e.g. W01, W02, ...) In the Writer Identification competition, predictions used the same format, and -1 for unknown writers. **Submission**

format: We had to submit submission_writer.csv with columns image_id, writer_id.

2.2 Challenges in the Dataset

The primary challenge in the datasets were similarity between signatures and their way of handwriting. There were 24 Unique writers and from naked eyes their signature looked exactly same. The most challenging part of this assignment was to predict -1 or unknown writers which were later added to the dataset.

There were no missing values in the dataset.The classes balanced among different writers. Only the unknown writer posed challenge since they were greater in number than the known writer.

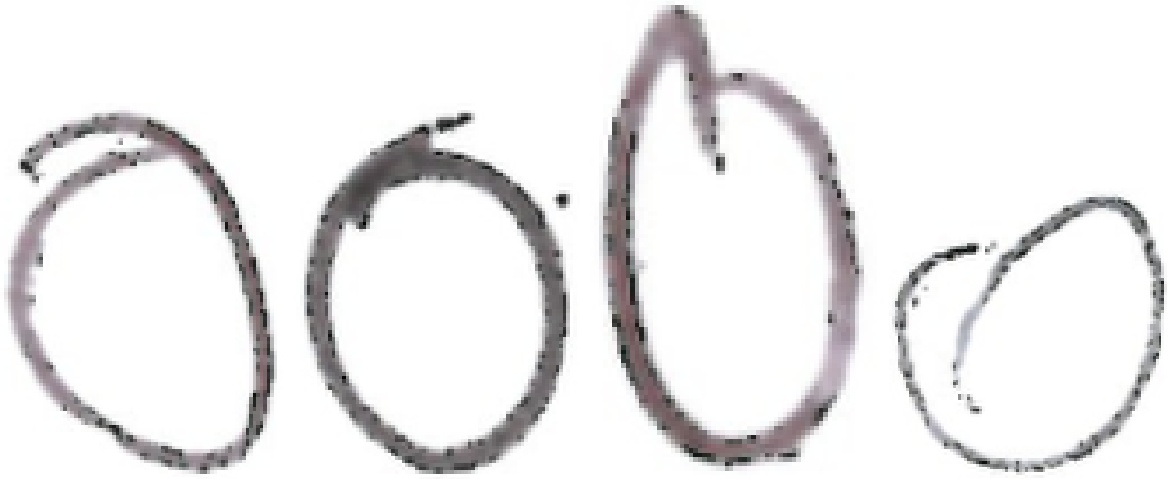

Figure 1: Images of Writer W27, W17, W01, W24 from left

2.3 Data Analysis

The dataset was divided into two files. First one being train.csv containing 23,850 rows and the other containing 16,400 rows. The total combined data were 40,250 unique images/records. Neither files contained any missing (Null) values. The dataset had following 4 columns **image_id, image_path, writer_id, pen_id** The dataset maps roughly 40,000 images to 44 known writers (plus unknown '-1' category) and 8 distinctly mapped pens. Because both datasets share the exact same columns and perfectly complement each other, we can safely merge them together for our machine learning or data analysis tasks.

3 Methodology / Approach

3.1 AC-GAN Discriminator

We trained an Auxiliary Classifier GAN(AC-GAN). The Generator continuously tried to created highly realistic fake handwriting for the 44 known writers. The Discriminator was forced to simultaneously solve two problems : first being predicting an image was Real Vs Fake, and classifying which of the 44 writers penned it.

Because we made the discriminator to constantly fight off a Generator that was actively trying to trick it with hyper-realistic fakes, it was forced to learn the absolute strictest, most microscopic biometric boundaries of what defined the 44 known writers. Standard classifiers never face this kind of pressure.

Finally we discarded the Generator, we took the battle-tested discriminator and deployed it as our standalone inference model.

When the kaggle test set is passed through this standalone Discriminator, it acted as an incredibly strict bouncer. By threshold analysis we determined the initial threshold to be 0.98. If the confidence was less than that we classified it as an unknown writer.

3.2 Kinematic Pressure Map & The Random Forest Bouncer

However, there was a problem as tracing the geometric shapes of a known writer's signature can be easy. Therefore using OpenCV we made a kinematic pressure Maps by analyzing stroke thickness (Distance Transforms) and ink intensity to measure raw pen pressure and stroke speed. This converted the 2D image into a flat , 1D biomechanical "fingerprint" that is physically impossible for a human forger to perfectly replicate.

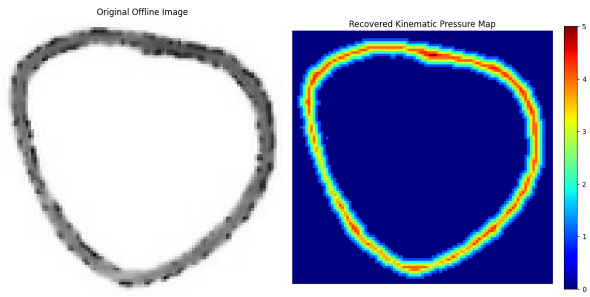

Figure 2: Kinematic Pressure Map

We fed these physical features into a Random Forest classifier to act as strict biometric bouncer. If a test signature had the correct visual shape but the wrong physical pressure profile, the Random Forest successfully flagged it as -1 Stranger.

3.3 Frequency Domain & Siamese Pyramidal Attention

Finally, we build a unified neural network capable of clustering known writers so tightly that stranger can be detected via strict distance threshold.

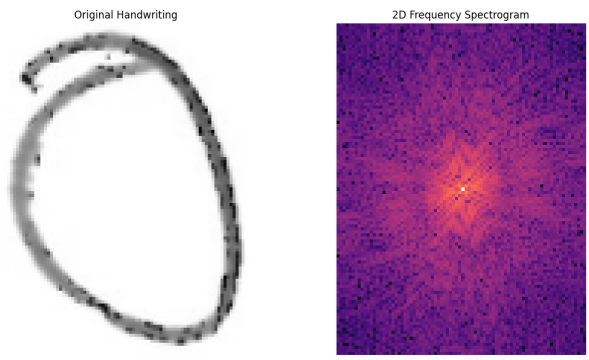

Figure 3: Frequency Domain Spectrogram

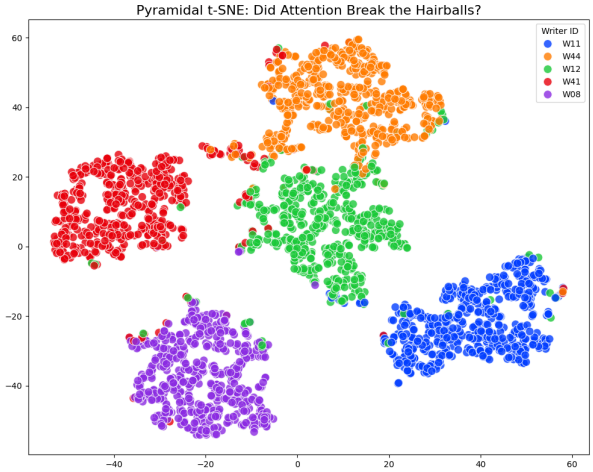

Figure 4: Islands Generated by Pyramidal Attention Network

We converted the handwriting images into 2D Fourier Spectrograms, destroying visual geometry to isolate pure biometric frequencies (slant, speed, and tremors). However there was a hurdle because Standard networks experienced partial mode collapse because high-frequency scanner noise overpowered the biometric signal of standard writers.Therefore we engineered a Multi-Scale Pyramidal Attention Network that simultaneously processed three distinct frequency lenses(Core 16x16, Mid 32x32, and Full 64x64).

Hence a dynamic attention layer automatically routed the correct frequencies for each writer, creating perfectly isolated 128-dimensional biometric islands and enabling our 0.85 anomaly threshold.

4 Result & Analysis

We achieved Rank 27 on private leaderboard with a Macro F1 Score of 0.42593 by reframing the open-set recognition challenge as an anomaly detection problem. Because standard classification networks suffer from severe overconfidence when forced to categorize unknown signatures, we shifted our focus to biometric physics and deployed an extracted AC-GAN Discriminator as our standalone inference model. By repurposing a network trained to catch hyper-realistic fakes, we estabilished an incredibly strict mathematically boundary around our 44 known writers.

This allowed the network to successfully recognize when a handwriting sample fell outside the true learned physical manifold, gracefully rejecting forgers while accurately classifying the known signatures.

Submission	Public Leaderoard	Private Leaderboard
submission_writer.csv	0.17670	0.15738
submission_writer(3).csv	0.43088	0.42593
submission_writer(8).csv	0.30406	0.27369
submission_ensemble.csv	0.30406	0.27313
submission_pyramidal	0.18896	0.17805

Table 1: Some of the Submission scores

5 Challenges

The main challenge in this competition was similarity of writers. Moreover even though we were able to segregate the individual writers from each other but it was quite difficult to separate themselves from the pool of unknown writers. This is clearly reflected in submission_pyramidal.csv from Table 1. Even though the model was able to generate the compact representation of each writer it was not able to isolate the unknown -1 writer efficiently.

6 Future Improvements

Future Improvements can be a neural graph approach where each individual writer can be represented as a node and their way of drawing a circle similarity as the edge between the two similar writer and their distinct on the node.

References

[1] Kaggle Competition Page :: Link

[2] ICDAR Page :: Click the Link to go to the Page

[3] GitHub Repository :: Link to Github